

58. Let the numbers be x and y such that $x > y$.
 Then, $x(x + y) = 204 \Rightarrow x^2 + xy = 204$...*(i)*
 and $y(x - y) = 35 \Rightarrow xy - y^2 = 35$...*(ii)*
 Subtracting *(ii)* from *(i)*, we get : $x^2 + y^2 = 169$.
 The only triplet satisfying this condition is (12, 5, 13).
 Thus, $x = 12$, $y = 5$.
59. Let the numbers be x and y .
 Then, $x + y = 20$ and $x - y = 8$.
 $\therefore x^2 - y^2 = (x + y)(x - y) = 20 \times 8 = 160$.
60. Let the numbers be x and y .
 Let $a - b = k$...*(i)*
 $a + b = 7k$...*(ii)*
 $ab = 24k$...*(iii)*
 Adding *(i)* and *(ii)*, we get : $2a = 8k$ or $a = 4k$.
 Putting $a = 4k$ in *(i)*, we get : $b = 3k$.
 Putting $a = 4k$ and $b = 3k$ in *(iii)*, we get : $4k \times 3k = 24k \Leftrightarrow 12k^2 = 24k \Leftrightarrow k = 2$.
 Hence, product of numbers = $24k = 24 \times 2 = 48$.
61. Let the numbers be x and y .
 Then, $xy = 120$ and $x^2 + y^2 = 289$.
 $\therefore (x + y)^2 = x^2 + y^2 + 2xy = 289 + 240 = 529$.
 $\therefore x + y = \sqrt{529} = 23$.
62. Let the numbers be x and y .
 Then, $xy = 45$ and $x^2 + y^2 = 106$.
 $(x + y) = \sqrt{(x^2 + y^2) + 2xy}$
 $= \sqrt{106 + 90} = \sqrt{196} \Rightarrow x + y = 14$...*(i)*
 $(x - y) = \sqrt{(x^2 + y^2) - 2xy}$
 $= \sqrt{106 - 90} = \sqrt{16} \Rightarrow x - y = 4$...*(ii)*
 Solving *(i)* and *(ii)*, we get : $x = 9$ and $y = 5$.
63. Let the numbers be x and y . Then,
 $x^2 + y^2 = 3341$...*(i)*
 and $x^2 - y^2 = 891$...*(ii)*
 Adding *(i)* and *(ii)*, we get :
 $2x^2 = 4232$ or $x^2 = 2116$ or $x = 46$.
 Subtracting *(ii)* from *(i)*, we get :
 $2y^2 = 2450$ or $y^2 = 1225$ or $y = 35$.
 So, the numbers are 35 and 46.
64. Let the numbers be x and $(x + 3)$. Then,
 $x^2 + (x + 3)^2 = 369$
 $\Leftrightarrow x^2 + x^2 + 9 + 6x = 369$
 $\Leftrightarrow 2x^2 + 6x - 360 = 0$
 $\Leftrightarrow x^2 + 3x - 180 = 0$
 $\Leftrightarrow (x + 15)(x - 12) = 0$
 $\Leftrightarrow x = 12$.
 So, the numbers are 12 and 15.
 $\therefore$ Required sum = $(12 + 15) = 27$.
65. Let the numbers be x and y .
 Then, $(x + y) = 22$ and $x^2 + y^2 = 404$.

$$\begin{aligned} \text{Now, } 2xy &= (x + y)^2 - (x^2 + y^2) = (22)^2 - 404 \\ &= 484 - 404 = 80 \Rightarrow xy = 40. \end{aligned}$$

66. Let the numbers be x and y .
 Then, $x^2 - y^2 = 256000$ and $x + y = 1000$.
 On dividing, we get : $x - y = 256$.
 Solving $x + y = 1000$ and $x - y = 256$,
 we get : $x = 628$ and $y = 372$.
67. Let the numbers be x and y .
 Then, $x^2 - y^2 = 63$ and $x - y = 3$.
 On dividing, we get : $x + y = 21$.
 Solving $x + y = 21$ and $x - y = 3$,
 we get : $x = 12$ and $y = 9$.
 $\therefore$ Larger number = 12.
68. Let $A = x$, $B = x + 2$, $C = x + 4$,
 $D = x + 6$ and $E = x + 8$.
 Then, $A + C = 146 \Rightarrow x + (x + 4) = 146$
 $\Rightarrow 2x = 142 \Rightarrow x = 71$.
 $\therefore E = x + 8 = 71 + 8 = 79$.
69. Let the six numbers be x , $x + 1$, $x + 2$, $x + 3$, $x + 4$
 and $x + 5$.
 Then, $x + (x + 1) + (x + 2) = 27 \Rightarrow 3x + 3 = 27$.
 Required sum = $(x + 3) + (x + 4) + (x + 5)$
 $= 3x + 12 = (3x + 3) + 9 = 27 + 9 = 36$.
70. Let the seven numbers be x , $x + 1$, $x + 2$, $x + 3$, $x + 4$,
 $x + 5$ and $x + 6$.
 Then, $x + (x + 1) + (x + 2) + (x + 3) + (x + 4) + (x + 5)$
 $+ (x + 6) = 175$
 $\Leftrightarrow 7x + 21 = 175 \Leftrightarrow 7x = 154$
 $\Leftrightarrow x = 22$.
 Required difference = $2(x + 6) - 3x = 12 - x$
 $= 12 - 22 = -10$.
71. Let the five numbers be x , $x + 2$, $x + 4$, $x + 6$ and $x + 8$.
 Then, $x + (x + 2) + (x + 4) + (x + 6) + (x + 8) = 575$
 $\Leftrightarrow 5x + 20 = 575 \Leftrightarrow 5x = 555 \Leftrightarrow x = 111$.
 $\therefore$ Required sum = $(x + 10) + (x + 12) + (x + 14)$
 $+ (x + 16) + (x + 18)$
 $= 5x + 70 = 5 \times 111 + 70 = 555 + 70 = 625$.
72. Let the three odd numbers be x , $x + 2$,
 $x + 4$ and the three even numbers be $x + 11$,
 $x + 13$ and $x + 15$.
 Then, $x + (x + 2) + (x + 4) + (x + 11) + (x + 13)$
 $+ (x + 15) = 231$
 $\Leftrightarrow 6x + 45 = 231 \Leftrightarrow 6x = 186 \Leftrightarrow x = 31$.
 $\therefore$ Required sum = $(x + 4) + (x + 15) = 2x + 19 = 2 \times 31$
 $+ 19 = 62 + 19 = 81$.
73. Let the three integers be x , $x + 2$ and $x + 4$.
 Then, $3x = 2(x + 4) + 3 \Leftrightarrow x = 11$.
 $\therefore$ Third integer = $x + 4 = 15$.
74. Let the four integers be x , $x + 2$, $x + 4$ and $x + 6$.
 Then, $x + (x + 2) + (x + 4) + (x + 6) = 1284$
 $\Leftrightarrow 4x = 1272 \Leftrightarrow x = 318$.
 $\therefore$ Greatest integer = $x + 6 = 324$.